AI, Your Brain, and the Tripos

Guidance on Effective and Responsible Use

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TL;DR: use this time to train your brain

The next one or two years are a golden opportunity

Make the most of it

You have time, difficult problems, other students and supervisors to help you develop your own understanding. Don't spend that opportunity prompting an AI to do the thinking you came here to practise.

- Human brains will still be needed, particularly those able to recognise and creatively develop the opportunities AI brings. AI can help build that understanding, but it can also compromise it.
- Ideally, supervision work should be entirely your own unaided work. Where it is not, make clear what assistance you used.
- AI is optional. The department supports your decision if you choose not to use it.

University policy and supervision guidance

University assessment rules; supervision guidance approved August 2026

Study and supervision

Appropriate AI use is permitted in personal study and formative work, within Tripos and course guidance. Supervision work is expected to be your own, normally produced without AI assistance. Legitimate accessibility support remains appropriate; discuss assistance openly.

Assessment

Unacknowledged AI-generated content submitted as your own in summative assessment is misconduct unless the brief explicitly permits it. Acknowledgement alone does not permit a prohibited use. CATAM prohibits AI-produced submitted code or write-up.

Expectations run both ways

Supervisors must not submit students' work to an AI tool's dataset. AI may help phrase their own CamCORS comments; feedback must not be entirely AI-generated.

University AI policy Supervision guidance, pp. 4–6 CATAM, section 5

An apparent paradox: AI research and unaided learning

The IoA and wider Cambridge astronomy are leaders in using AI for research

New research capability

We use AI to extend what we can do in research. This is a substantial opportunity for astronomy, and we want people who can creatively recognise and develop it.

Developing the people who will use it

Researchers bring years of learning and experience to deciding what to ask, which directions to pursue and what the answers mean. Developing that understanding is part of what you are here to do.

The same reason for both positions

Our enthusiasm for AI in research makes your own intellectual development more important. We want you to be able to shape what these tools make possible.

Hiranya V. Peiris, *Large language models are not the problem*, [2604.22071]

Natalie B. Hogg, *Agentic research is oxymoronic*, [2608.31161]

Context: A Departmental Investigation

An Internal Experiment

An AI model (o3 from OpenAI) sat the entire Part II Astrophysics written examination at the same time as students in 2025.

Toby Lovick transcribed the model's responses into exam booklets. These were marked blind alongside student scripts by the official examiners.



Results and Interpretation

The AI achieved a first-class mark, placing well above typical student performance.

This was a departmental experiment with an experienced user. Since then, mathematical capabilities have moved substantially beyond this result. Model, tools and context still matter.

AI is now producing new mathematics

OpenAI's September and October 2026 announcements

Beyond examination questions

- September: a proposed resolution of the Navier–Stokes Millennium Problem, with a proof and Lean formalisation.
- The construction gives finite-time blow-up with smooth external forcing. This is not a general solution of fluid dynamics.
- October: a broad release of new mathematical results, with many proofs formalised in Lean.

A research effort with human direction

Researchers selected problems, redirected resources after intermediate results and combined insights between agent groups. These internal systems demonstrate extraordinary problem-solving; they do not establish autonomous research direction.

OpenAI, 8 September 2026 OpenAI, 6 October 2026

Implications for Your Studies

Definition

This technology presents both an opportunity to enhance learning and a risk to academic development. This talk provides guidance on its responsible use. **AI use is optional: the department supports your decision if you choose not to use it.**

Acceptable Use: AI as a Tool



To augment your thinking, deepen understanding, and explore concepts more effectively.

Identified Risk: AI as a Trap



To bypass learning, avoid difficult problems, and outsource intellectual effort.

The Unique Opportunity

These next one or two years are a golden opportunity to train your brain. Working through problems yourself builds the understanding and judgement you will take into an AI-rich world. Make use of the people and teaching around you.

Existing Safeguards in the Cambridge System

The structure of the Cambridge degree provides inherent safeguards against the misuse of these tools.

1. The Supervision System

- Students and supervisors **test understanding together** through discussion, with both explaining their reasoning and responding to questions.
- AI can help develop understanding, but a submitted solution does not by itself establish understanding.

2. The Examination System

- Your degree is assessed almost entirely by **closed-book, invigilated examinations**.
- This assessment format is a robust test of unaided individual knowledge.

Note

You may be asked to explain or reproduce your work in a viva or monitored environment, particularly for coursework, projects, and lab reports.
Using AI for supervision work can mask difficulties that supervisors need to identify to provide proper support.

The Central Assessment: The Tripos Examination

A Note of Caution

- Your degree is primarily assessed via closed-book, invigilated examinations.
- Using AI to bypass the work on examples sheets can leave you without the independent problem-solving abilities that the Tripos examines.
- The concern includes students using AI earnestly and mistaking a solution they can follow for one they could construct.



Consequences of Misuse

Misuse in summative work will be reported. Sanctions range from mark penalties to failure of the year, depending on the severity of the academic misconduct.

Cognitive Risk: The 'Illusion of Competence'

A key pedagogical risk is mistaking recognition for understanding.

1. You encounter a difficult problem and struggle to begin.
2. You consult an AI, which provides a correct, well-explained solution.
3. You read the solution and find it easy to follow, leading to the thought, *"That's straightforward, I could have done that."*



This bypasses a critical stage of learning

Working through difficulty is part of developing independent understanding. Research on "desirable difficulties" distinguishes immediate fluency from lasting learning (Bjork & Bjork, 2011).

Studies of AI-assisted essay writing ([2506.08872]) and self-reported critical thinking at work (Lee et al., 2025) raise concerns about cognitive engagement. These are not direct tests of Tripos learning.

Exploitation vs exploration in research

An analogy from optimisation and reinforcement learning

Exploitation

Pursue a promising direction using what is already known. AI brings unprecedented speed and breadth, including original, difficult mathematics. This need not be routine work.

Exploration

Seek new directions and decide which questions are worth pursuing. Humans still guide the overall research programme, including at frontier AI labs. Autonomous research direction has not been established.

Search within a specified problem can itself involve exploration. The distinction here concerns direction of the research programme; current "auto-research" remains bounded and human-directed.

The educational difficulty

That judgement includes choosing the physical model, deciding when an approximation is valid and interpreting an unexpected result. How do we develop it when the calculations and coding through which we learnt it can be automated? This is not settled.

Sutton & Barto, Reinforcement Learning, chapters 1–2

The frontier is still jagged

Both capabilities and failure modes have moved

- Frontier models have substantial capabilities. In some circumstances I would trust their answer more than some published papers.
- Intuitions based on tools used a year or more ago can now be badly out of date. The frontier has moved, but serious errors remain.

More than "hallucination"

In my own use, models can be duplicitous about the work they have done: choosing to read only a fraction of the context, concealing that shortcut and presenting a confident answer that is substantively misleading or outright wrong.

Judgement

Check what was actually read and tested, as well as the assumptions and reasoning. A confident answer need not address the full problem, or accurately describe how it was obtained.

Write your Part III project yourself

Developing your own scientific voice is part of the work

Authorship and the published requirements

The Part III report must rest on your own ideas, analysis, conclusions and understanding. AI assistance must be identified and may supplement, not substitute for, your work. PhD theses likewise require an own-work declaration and identification of AI-generated content.

My recommendation

Write the material yourself. If you use AI on the prose, treat it as a souped-up grammar checker, not the author. Follow the project rules and agree your use with your supervisor.

A choice my PhD students make

My PhD students choose to write their theses themselves to develop their own writer's voice. You will have few opportunities, perhaps never another like this, to spend sustained time learning to write.

Part III booklet 2026–27, p. 7 Research thesis declaration

Recommended Usage Framework

If you choose to use AI, the goal should be to enhance, not replace, the learning process.

Important

Ideally, prepare supervision work entirely unaided; disclose and agree any assistance with your supervisor. When stuck, first consult textbooks, lecture notes, supervisors and other students. For revision and other permitted uses, AI can help with explanations, questions, hints and feedback. Cross-check its suggestions against these sources. Keep practising unaided; AI remains optional.

Ineffective/Risky Use ×

- "Give me the solution to Tripos 2022 Paper 3 Question 4."
- "Write my essay on the formation of the solar system."
- "Write corrected code for my CATAM submission." (Forbidden)

Effective Use ✓

- "Explain this concept from my notes in a different way."
- "Mark my attempt at this question and provide feedback."
- "Generate new practice questions on a topic I'm struggling with."

Effective Use Case 1: Conceptual Exploration

Gaining different perspectives on lecture material.

In our lab: AI that slows you down

Natalie Hogg's `read-paper` skill makes you commit to an interpretation, then read and answer questions before it explains. It checks recall and limits detours into derivations.

Here AI is deliberately slowing us down to help us understand.

Prompt: Generating Questions From Your Notes

```
Here are my summary notes on General Relativity: [paste notes]. Generate three
Tripos-style questions based on this material. One should be a short, bookwork-style
question and two should be longer, problem-solving questions.
```

Consult textbook problem sets first for practice material. AI also offers a patient space for questions you might hesitate to ask in a group. Check generated questions against notes, textbooks and discussion with your supervisor.

Data Protection and Copyright

Check permission and assessment rules before uploading materials, and how the tool stores and uses your data.

Effective Use Case 2: Feedback on Formative Work

Revision after a supervision, or other use agreed with your supervisor.

Workflow

1. Attempt questions independently, under exam conditions. Use your attempt for feedback after supervision, or in another agreed learning context.
2. Take a clear photo of your handwritten work.
3. If permitted, upload it to a suitable multimodal AI tool.
4. Ask for feedback or diagnostic questions, not just a corrected answer.

Important Caveats

Be mindful of copyright and data protection when uploading. Models can miss errors or invent objections. Cross-check against textbooks and lecture notes; take unresolved disagreements to your supervisor.

Prompt: Requesting Feedback

```
You are a Cambridge supervisor. The attached image is my attempt at a quantum mechanics problem. Please:
```

- `Mark my answer, identifying mathematical errors.`
- `Critique my physical reasoning and the validity of my assumptions. Ask questions to diagnose what I have misunderstood.`
- `Where I got stuck, give a hint without completing the solution.`

Effective Use Case 3: Targeted Exam Practice

Generating additional practice problems for topics you find difficult.

Prompt: Generating New Exam Questions

```
I am a Part III astrophysics student preparing for examinations. I need more
practice on magnetohydrodynamics. Below are three examples of past Tripos
questions on this topic. [Paste 2-3 past paper questions here] Generate two new,
unseen questions of a similar style and difficulty that test the same core
principles. Then, provide a brief mark scheme for each.
```

Quality Check Required

Check for novelty, correctness, and appropriate scope. Refine mark schemes first with textbooks and collaboration with supervisors and cohort members.

Pedagogical Value

This requires active engagement with past papers. The primary task—solving the problems—remains your own work. Revision using past papers and textbooks alone is perfectly sufficient; AI is optional.

Summary

Key Points

- AI can solve difficult open problems, yet give confident, misleading answers and conceal incomplete work. The frontier is jagged and moving.
- A strong answer from a model does not establish your own understanding. Use assistance to develop the abilities you need yourself.

Primary Risks

- Following a solution can feel like being able to construct it. Preserve independent attempts and unaided practice.
- AI-produced CATAM submissions breach the course rules.

Recommendations

If using AI:

- Treat it as a Socratic tutor, not an answer service.
- Use explanations, feedback and questions to develop your own reasoning.
- Use it to generate new challenges, not to avoid existing ones.

The primary goal is to develop your own analytical judgement. Your brain is what is being examined.

Questions?